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Ariel Slepyan

Contact Information

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Summary

ICoN Postdoctoral Fellow at MIT working on scalable tactile sensing, sensorimotor control, and robot learning. My PhD established compressive sampling as a general design principle for tactile skin, breaking the traditional tradeoff between density, speed, wiring, and area. My adaptive sampling method reads standard passive arrays 18X faster than raster scanning with no added sensor-level circuitry. I lead OpenTac, an open-source ecosystem for designing custom tactile skins. My work has spun out 2 hardware companies, produced 7 patents and invention disclosures, and received $690K in research funding.

Education

08/2021 – 06/2026	PhD in Electrical and Computer Engineering Johns Hopkins University — Baltimore, MD, USA Advisor: Professor Nitish Thakor Thesis: "*Scalable Large-Area, High-Density Tactile Sensing through Compressive Sampling*"
08/2020 – 08/2021	M.S.E in Biomedical Engineering Johns Hopkins University — Baltimore, MD, USA Advisor: Professor Nitish Thakor Dean's Master's Fellowship Thesis: "*Scalable Tactile Sensing E-Skins Through Spatial Frequency Encoding*"
08/2016 – 05/2020	B.S. in Biomedical Engineering Johns Hopkins University — Baltimore, MD, USA Bloomberg Scholarship

Experience

07/2026 –	ICoN Postdoctoral Fellow CSAIL and BCS MIT — Cambridge, MA, USA Advisors: Professors Pulkit Agrawal and Nidhi Seethapathi --*Scalable tactile sensing to learn from human data and improve robots*
03/2026 – 06/2026	Research Scientist Intern Reality Labs Research, Robotics Meta — Redmond, WA, USA Part-time remote extension, 06/2026 – 09/2026 --*Focusing on scalable tactile sensing in dexterous robotic hands*
06/2020 – 06/2026	Graduate Student Neuroengineering & Biomedical Instrumentation Lab Johns Hopkins University — Baltimore, MD, USA --*Focusing on scalable touch sensing in robotics and wearables*
01/2024 – 07/2026	Co-Founder Navonic — Baltimore, MD, USA --*Wearable sensors for motion tracking and XR*
01/2023 – 12/2024	Lead Hardware Engineer CurveAssure Spine — Baltimore, MD, USA --*Lead hardware development of a wearable spine monitor*
Summer 2019	Summer Researcher

	Singapore Institute for Neurotechnology *--Developed RFID-based wireless touch sensors*	Singapore
Summer 2018	Summer Researcher (Visiting Scholar) Interuniversity Microelectronics Centre (IMEC) *--Built microfluidic droplet sorter using DEP in silicon chip*	Leuven, Belgium
01/2017 – 01/2019	Undergraduate Researcher BioMEMS Lab Johns Hopkins University *--Built microfluidic droplet-based platform for measuring enzyme kinetics*	Baltimore, MD, USA
Summer 2014/15	Research Intern Groisman Lab University of California, San Diego *--Build PDMS chip to investigate thermotaxis of E. coli*	La Jolla, CA, USA

Awards & Honors

2025	Winner of Best Demo Award at NER 2025
2025	Winner of Best Live Demo Award at ISCAS 2025 (1st / 34 demos)
2025	Winner of Pava Center Innovation & Entrepreneurship Award (Navonic, $15k)
2024	Willard & Marilyn Sweetser ARCS Foundation Scholar ($15k)
2024	Winner of JHU Fuel Cohort Prize (Navonic-Evoked Haptics, $8k)
2024	Winner of WSE Excellence in Teaching, Advising, and Mentoring Award
2023	Winner of JHU President's Venture Fellowship (CurveAssure, $100,000)
2023	Grand Prize Winner of JHU Makerspace Design Challenge
2023	Finalist for IEEE World Haptics Student Innovation Challenge (P1)
2022	Invitee to the Telluride Neuromorphic Cognition Engineering Workshop (1 of 30 international)
2022	Grand Prize Winner of Johns Hopkins ECE Design Day
2022	Finalist in Johns Hopkins Healthcare Design Competition ($5,000)
2020	Recipient of Dean's Master's Fellowship (1/2 Tuition Scholarship)
2019	Finalist for FastForward Summer Award ($10,000)
2018	Winner of Johns Hopkins Spark Grant ($1,000)
2017	2nd place overall winner at MedHacks 2017
2017	Winner of Wolfram Award and Contrary Capital Prize at MedHacks 2017
2016	Recipient of Michael R. Bloomberg Scholarship (Full Tuition 4-year Scholarship)
2016	2nd place in Microbiology at Intel ISEF 2016 (International Science and Engineering Fair)
2016	1st place in Engineering at the New York State Science and Engineering Fair
2015	1st place in Materials Science at the New York State Science and Engineering Fair

Research Grants Awarded ($690k)

- Analog Devices 2026 ($300,000) – "Learning kHz Muscle-Memory Control Loops for Dexterous Robotic Hands" (PI: Prof. Pulkit Agrawal)
- a2 Pilot Awards: Artificial Intelligence and Technology Collaboratory (AITC) 2025 ($200,000) – "Re-Kinesis: Wearable Gait Analysis Laboratory for Aging Populations"
- Accelerated Translational Incubator Pilot Program (ATIP) 2025 ($50,000) – "Re-Kinesis Twin: Democratizing Gait Analysis in Rehabilitation with e-Dermis and AI-Powered Digital Twin."
- TEDCO MII Technology Assessment Award 2024 ($115,000) – "High-Density Tactile Sensor Array with a Single Output Wire for Medical Rehabilitation". (#1 score in cohort 4.167 / 5)
- Space@Hopkins Seed Grant Program 2022 ($25,000) – "Self-Powered, Electronics-Free Tactile Sensors Immune to the Hazards of Cosmic Radiation"

Patents and Invention Disclosures

1. "Systems and Methods for Learned Adaptive Measurement Selection Using Greedy-Oracle Supervision" (JHU Tech ID #C19459)
2. "Systems and Methods for Multi-Axis Force Extraction from High-Resolution Tactile Arrays" (JHU Tech ID #C19151)

3. "Wearable Plantar Sensor-Based System for Lower Limb Biomechanics Reconstruction" (JHU Tech ID #C18700)
   Provisional Patent Application Number = 63/992,527
4. "Scalable Distributed Tactile Sensors Without Integrated Circuits Using Resonance Multiplexed Piezoelectric Sensors" (JHU Tech ID #C18042)
   Provisional Patent Application Number = 63/682,440
5. "Method and Apparatus of a High-Density Non-Invasive Neural Stimulator" (JHU Tech ID #C18182)
   Provisional Patent Application Number = 63/655,671
6. "Method of Scalable Sensor Arrays Through Row Column Compressive Sensing" (JHU Tech ID #C18041)
   Provisional Patent Application Number = 63/653,517
7. "Scalable, Event-Based Sensing Using Wireless Sensor Elements Embedded in Flexible Elastomer" (JHU Tech ID #C16118)
   Patent Number = 18/553,717, Published 2022-10-06

Research Papers

Preprints and Under Review

1. **A. Slepyan**, G. C. R. Bethala, J. Gu, A. Dangle, Y. Zhou, H. Fu, J. Sim, T. Basu, X. Yin, F. Karim, G. Hager, H. Bharadhwaj, A. Bhattad, K. M. Jatavallabhula, and N. Thakor, "OpenTac: An Open Ecosystem for Customizable, Scalable, and Transferable 3D Tactile Sensing", *arXiv preprint, 2026,* https://arxiv.org/abs/2610.31661
   Open-source hardware and software - https://opentac.github.io
   Also presented at 2026 CoRL Workshop on Everything Beneath the Policy, 2026 CoRL Workshop on Dexterous Manipulation, IEEE Haptics Symposium 2026 (Open-Tactile CAD)
2. W. Lu*, **A. Slepyan***, P. Perkins, A. Ascani Orsini, K. Quinn, and N. Thakor, "Scalable high-density tactile arenas for measuring ground-contact forces in freely moving animals", *bioRxiv preprint, 2026,* https://www.biorxiv.org/content/10.64898/2026.10.22.760735v1
   Open-source hardware, software, and data - https://tactilearena.github.io
3. **A. Slepyan***, D. Li*, A. Aug, S. Sankar, T. Tran, and N. Thakor, "Adaptive Compressive Tactile Subsampling: Enabling High Spatiotemporal Resolution in Scalable Robotic Skin", *Under Review,* https://arxiv.org/abs/2410.13847
   Also presented at 2024 Workshop on Neuromorphic Principles in Biomedicine and Healthcare

Published

4. **A. Slepyan,** Y. Luo, E. Levenshus, et al, "Analog Dual-Tone Multi-Frequency Signaling for Scalable Tactile Skin with Automatic Data Compression and a Single-Wire Output", IEEE Sensors Journal, https://ieeexplore.ieee.org/abstract/document/11516513

5. **A. Slepyan*,** R. Zhang*, L. Xing*, and N. Thakor, "Single-Pixel Tactile Skin via Compressive Sampling", Communications Engineering (Nature Portfolio), https://www.nature.com/articles/s44172-026-00697-2
   Also presented at 2025 Northeast Robotics Colloquium

6. S. Bellow, **A. Slepyan**, J. Chen, et al, "Towards the Estimation of Lower Limb Flexion Angles using High-Density Plantar Pressure Sensing", 2026 48th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC). *Accepted*

7. **A. Slepyan**, J. Chen, and N. Thakor, "Scalable Tactile Sensing Skins: Wiring and Data Management" *Proceedings of the IEEE*. 2026. https://ieeexplore.ieee.org/document/11393898

8. D. Poppert*, **A. Slepyan***, N. Thakor, T. Tran, "ReLACS: Responsive Learned Adaptive Compressive Subsampling for Efficient Readout of Large-Area Tactile Skins", 2026 Data Compression Conference (DCC). https://ieeexplore.ieee.org/abstract/document/11510445

9. K. Cheng, M. Wang, **A. Slepyan**, and N. Thakor, "Biomimetic Fingertips: Friction Modulation Through Surface-Responsive Texture Triggered by Micro-Vibration Sensing", *IEEE Sensors Letters*, 2025. https://ieeexplore.ieee.org/document/11178177

10. V. Chari, M. Iskarous, D. Li, **A. Slepyan**, N. Thakor, and J. Downey, "Neuromorphic Compression of Tactile Data with Spiking Neural Networks", 12th International IEEE/EMBS Conference on Neural Engineering, 2025. *Accepted.*

11. A. Pimpalkar, **A. Slepyan**, and N. Thakor, "Vibrations At First Contact Encode Object Stiffness Before Grasp Completion", *IEEE Sensors Letters*. 2025 https://ieeexplore.ieee.org/document/11060835

12. M. Iskarous, Z. Chaudhry, F. Li, S. Bello, S. Sankar, **A. Slepyan**, et al, "Invariant neuromorphic representations of tactile stimuli improve robustness of a real-time texture classification system" *Advanced Intelligent Systems.* 2025, https://advanced.onlinelibrary.wiley.com/doi/10.1002/aisy.202401078

13. S. Sankar, W. Cheng, J. Zhang, **A. Slepyan**, et al "A natural biomimetic prosthetic hand with neuromorphic tactile sensing for precise and compliant grasping" *Science Advances.* 2025. https://www.science.org/doi/10.1126/sciadv.adr9300
    Featured in JHU HUB - https://hub.jhu.edu/2025/03/05/prosthetic-robotic-hand/

14. Y. Angkanapiwat, **A. Slepyan**, and N. Thakor, "SensoPatch: a Reconfigurable Haptic Feedback with High-Density Tactile Sensing Glove," 2024 IEEE Biomedical Circuits and Systems Conference (BioCAS), 2024. https://ieeexplore.ieee.org/document/10798282

15. **A. Slepyan**, M. Zakariaie, T. Tran, and N. Thakor "Wavelet Transforms Significantly Sparsify and Compress Tactile Interactions" Sensors. 2024. https://www.mdpi.com/1424-8220/24/13/4243

16. S. Wang, K. Quinn, **A. Slepyan**, et al, "Channel selection and wavelet transformation-based data compression preserve motor unit information" 2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC). 2024. https://ieeexplore.ieee.org/document/10782539

17. **A. Slepyan***, S. Krishnan*, T. Li and N. Thakor, "A Multi-Channel, Low-Voltage, High-Frequency Programmable Electrical Stimulator for Sensory Feedback," 2023 IEEE Biomedical Circuits and Systems Conference (BioCAS), 2023. https://ieeexplore.ieee.org/document/10388769

18. Z. Ou, Y. Guo, P. Gharibani, **A. Slepyan**, et al "Time-frequency analysis of somatosensory evoked high-frequency (600 Hz) oscillations as an early indicator of arousal recovery after hypoxic-ischemic brain injury" Brain Sciences. 2022. https://www.mdpi.com/2076-3425/13/1/2

19. Y. Tian, **A. Slepyan**, et al, "Real-Time, Dynamic Sensory Feedback Using Neuromorphic Tactile Signals and Transcutaneous Electrical Nerve Stimulation" 2022 IEEE Biomedical Circuits and Systems Conference (BioCAS), 2022. https://ieeexplore.ieee.org/document/9948609

20. S. Sankar, **A. Slepyan**, et al, "Flexible Multilayer Tactile Sensor on a Soft Robotic Fingertip" 2022 IEEE Sensors, 2022. https://ieeexplore.ieee.org/document/9967059

21. A. Aug, **A. Slepyan**, E. Levenshus, N. Thakor, "Haptic Touch: A retrofittable tactile sensing glove and haptic feedback armband for scalable and robust tactile sensory feedback", 2022 9th IEEE International Conference on Biomedical Robotics and Biomechatronics (BioRob), Seoul, Korea, 2022. https://ieeexplore.ieee.org/document/9925475

22. **A. Slepyan**, S. Sankar, and N. Thakor, "Texture Discrimination Using a Neuromimetic Asynchronous Flexible Tactile Sensor Array with Spatial Frequency Encoding", 10th International IEEE/EMBS Conference on Neural Engineering, 2021 https://ieeexplore.ieee.org/document/9441136

23. **A. Slepyan** and N. Thakor, *"Towards scalable soft e-skin:* Flexible event-based tactile-sensors using wireless sensor elements embedded in soft elastomer", 2020 8th IEEE International Conference on Biomedical Robotics and Biomechatronics (BioRob), New York, 2020 https://ieeexplore.ieee.org/document/9224353

Research Abstracts / Short Papers

1. **A. Slepyan**, G. C. R. Bethala, J. Gu, A. Dangle, Y. Zhou, H. Fu, J. Sim, T. Basu, X. Yin, F. Karim, G. Hager, A. Bhattad, H. Bharadhwaj, K. M. Jatavallabhula, and N. Thakor, "Scalable 3D Force-Sensing Tactile Skins for Robot Learning" 2026 Conference on Robot Learning (CoRL), Demo Track. Accepted.

2. **A. Slepyan**, A. Orsini, K. Quinn, and N. Thakor "A High-Density Tactile Arena for Neuroengineering Research: Continuous Gait Tracking in Freely Behaving Rodents" 2025 IEEE EMBS 12th Annual International Conference on Neural Engineering (NER 2025).
   Winner of Best Demo Award

3. **A. Slepyan***, D. Li*, T. Tran, and N. Thakor, "Live Demonstration: Compressive Subsampling for High-Speed Large-Area Tactile Sensing" 2025 IEEE International Symposium on Circuits and Systems (ISCAS). 2025. https://ieeexplore.ieee.org/abstract/document/11043348
   Winner of Best Live Demo Award

4. **A. Slepyan** and N. Thakor "PAS: Probabilistic Adaptive Subsampling for Scalable Tactile Skins" 2025 ICRA ViTac Workshop. https://shanluo.github.io/ViTacWorkshops/content/ViTac2025_Paper_10.pdf

5. S. Bello, **A. Slepyan**, J. Chen, and N. Thakor "Predicting Kinematics from High-Density Plantar Pressure Sensing" 2025 ICRA ViTac Workshop. https://shanluo.github.io/ViTacWorkshops/content/ViTac2025_Paper_09.pdf

6. **A. Slepyan***, D. Li*, T. Tran, and N. Thakor, "Compressive Subsampling for Scalable Tactile Skin" 2025 Data Compression Conference (DCC). 2025. https://ieeexplore.ieee.org/document/10992472

Workshop Presentations

1. L. Feng, **A. Slepyan**, K. Murthy, and N. Thakor, "LC 3-Axis: An Accurate Three-Axis Force Sensor Assembled with Readily Available Off-the-Shelf Components with No Designed Compliant Element" 2026 IROS Workshop on Sensors and Actuators for Dexterous Manipulation. https://lucasfeng7.github.io/lc-3-axis/

2. **A. Slepyan**, M. Iskarous, S. Sankar, and N. Thakor, "Scalable, Biomimetic Sensory Solutions for Dexterous Robotics Hands" 2021 NRI & FRR Principal Investigators' Meeting, 2021

3. **A. Slepyan**, R. Acharya, A. Silva, D. Kumar, and N. Thakor, "A Biomimetic Soft Finger for Palpation Applications", Do Good Robotics Symposium, Maryland, 2019

4. **A. Slepyan**, N. Ribeiro, A. Saad-Eldin, A. Blakney, "Rapid development of paper-based microfluidic devices using crayons and coffee filters", Baltimore Innovation Week Science Conference, 2017

Professional Activities

- Reviewer for IEEE Robotics and Automation Letters (RA-L)
- Reviewer for IEEE Transactions on Mechatronics (TMECH)
- Reviewer for IEEE Transactions on Very Large Scale Integration Systems (TVLSI)
- Reviewer for 2027 IEEE International Conference on Robotics and Automation (ICRA)
- Reviewer for 2027 Robotics: Science and Systems (RSS)
- Reviewer for 2026 IEEE International Symposium on Circuits and Systems (ISCAS)
- Reviewer for 2026, 2025 IEEE International Conference on Intelligent Robots and Systems (IROS)
- Reviewer for 2025 IEEE World Haptics Conference (WHC)
- Reviewer for 2025, 2024, 2022 IEEE International Conference on Electronics, Circuits & Systems (ICECS)
- Reviewer for 2024, 2022 IEEE International Conference on Biomedical Robotics and Biomechatronics (BioRob)

Workshops Organized

- Homanga Bharadhwaj, Irmak Guzey, **Ariel Slepyan**, Haozhi Qi, Wenzhen Yuan, and Gregory Hager: **"Scalable Tactile Sensing for Dexterous Manipulation"**, 2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).

Teaching Lead of Original Courses (110+ students)

EN.520.299	PCB Design and Microcontroller Programming	Winter 2023 – 2025
	Student feedback – "Arik is extremely helpful, and willing to put in the extra time for both students who are struggling and those who want to go above and beyond. 10/10 instructor!"	
EN.580.113	Prosthesis Instrumentation	Winter 2023
	HUB Article – https://hub.jhu.edu/2023/02/06/intersession-prosthesis-instrumentation/	

Teaching Assistantships (17 classes)

EN.580.471	Principles of the Design of Biomedical Instrumentation	Fall 2020 – 2025
EN.520.448	Advanced Electronics Design Lab	Spring 2023 – 2024
EN.580.571	Honors Instrumentation	Spring 2021 – 2025
EN.580.456	Introduction to Rehabilitation Engineering	Fall 2022 – 2023
n/a	BME Design Studio TA	2021 – 2022
EN.580.457	Rehabilitation Engineering: Design Lab	Spring 2021
EN.580.477	Biomedical Data Science Lab	Fall 2019

Leadership Experience

2023 – 2025	Co-President of ECE Graduate Student Association
2023	ECE representative for the Graduate Student Organization (GRO)

Student Mentorship (40+ students), (15 published *)

Master's Students: Dian Li* (2024 – 2025), Kai Cheng* (2024 – 2025), Anway Pimpalkar* (2024 – 2025), Dheeraj Gudluru (2024), Diego Gomez (2024), Junjun Chen* (2024 –), Priyanka Fernandes (2023 – 2024), Tianao Li* (2023 – 2025), Siddharth Krishnan* (2022 – 2023), Michael Zakariaie* (2022 – 2023), Arnab Chatterjee (2022), Yucheng "Jacky" Tian* (2022)

Undergraduate Students: Colman Barth (2025), Joshua Pike (2025), Ryan McGovern* (2025), Jacobo Tello (2025 –), Colm Doley (2025 –), Ian Tran (2025), Ashley Luo* (2024 – 2025), Keya Agrawal (2024), Rudy Zhang* (2024 – 2025), Amanda Butler (2024), Mathew Schricker (2023 – 2024), Laura Xing* (2023 – 2025), Yanisa "Belle" Angkanapiwat* (2022 – 2025), Guangyan "Molly" Li (2022 – 2023), Eli Levenshus* (2021 – 2023), Aidan Aug* (2021 – 2023), Dylan Zhu (2022 – 2023), Neeti Prasad (2022), Aryaman Shodhan (2021), Martin Prados de Haro (2021)

High School Students: El Donald (2024), Srinitha Kondapaneni (2023), Ian Tran (2023), Upanshu Bajaj, Shriya Sane, and Kimaya Basu (2022 – 2023), Pratham Mathapati, Pranavaa Elangovan, Sahil Mada (2020 – 2021),

Consulting Work

- Hardware Design for Somnair
  Developed a multi-channel non-invasive neurostimulator (electrode array + stimulator)
- PCB Design for OmniSense Surgical, Inc
  Developed a "smart retractor" providing real-time surgical guidance with a focus on mastectomy procedures and plastic & reconstructive surgery. Startup spun out of the Texas Biodesign program.